Survey Methods
4.5 – Data management

Dr Annibale Cois, MEng, MPH, PhD
Division of Health Systems and Public Health
acois@sun.ac.za

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Outline

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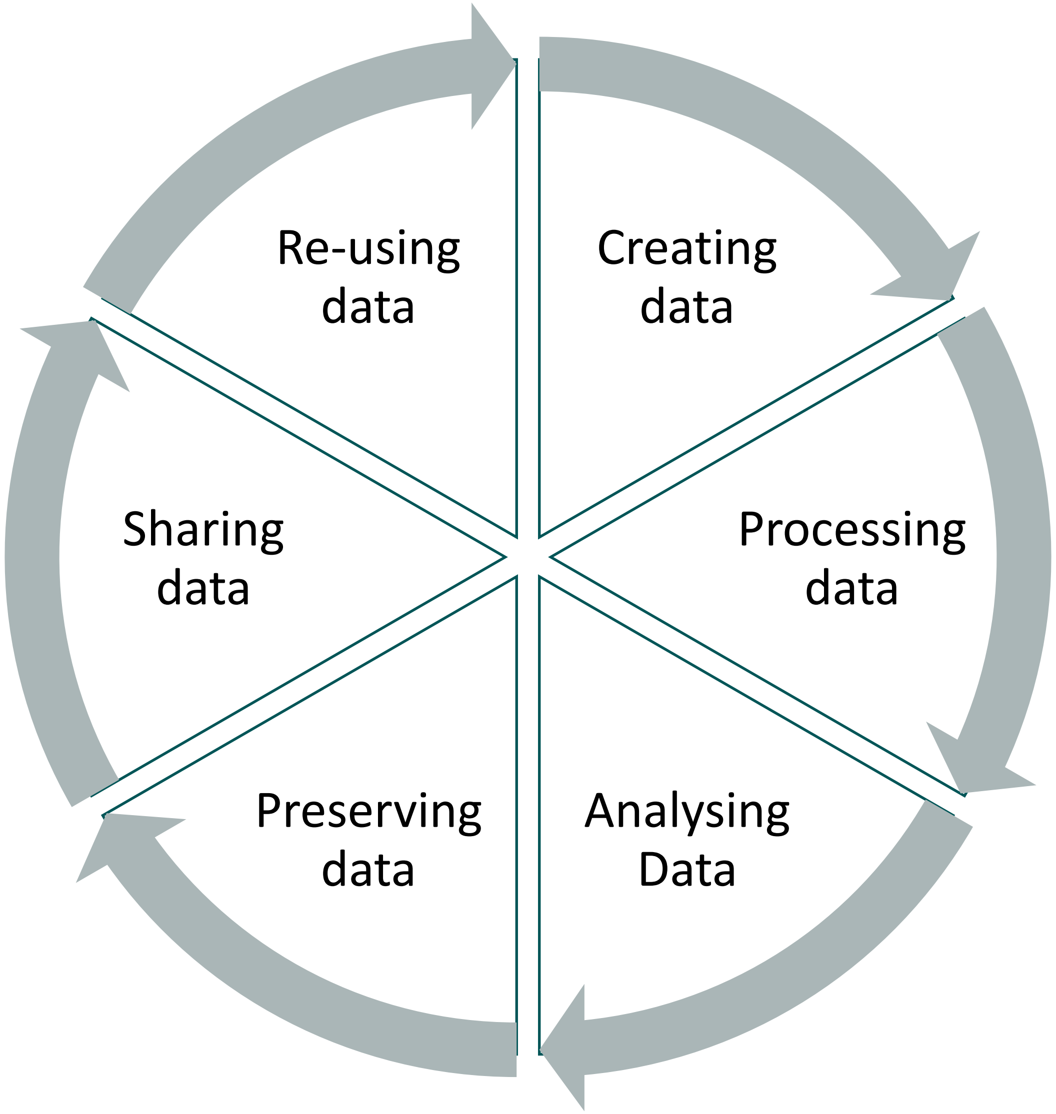
- Principles
- Checking, correcting, documenting
- Data management plan
- Data keeping and sharing

(Research) Data management?

Research data management (RDM) refers to the organisation, storage and preservation of data created during a research project. It covers initial planning, day-to-day processes and long-term archiving and sharing.

Data Lifecycle

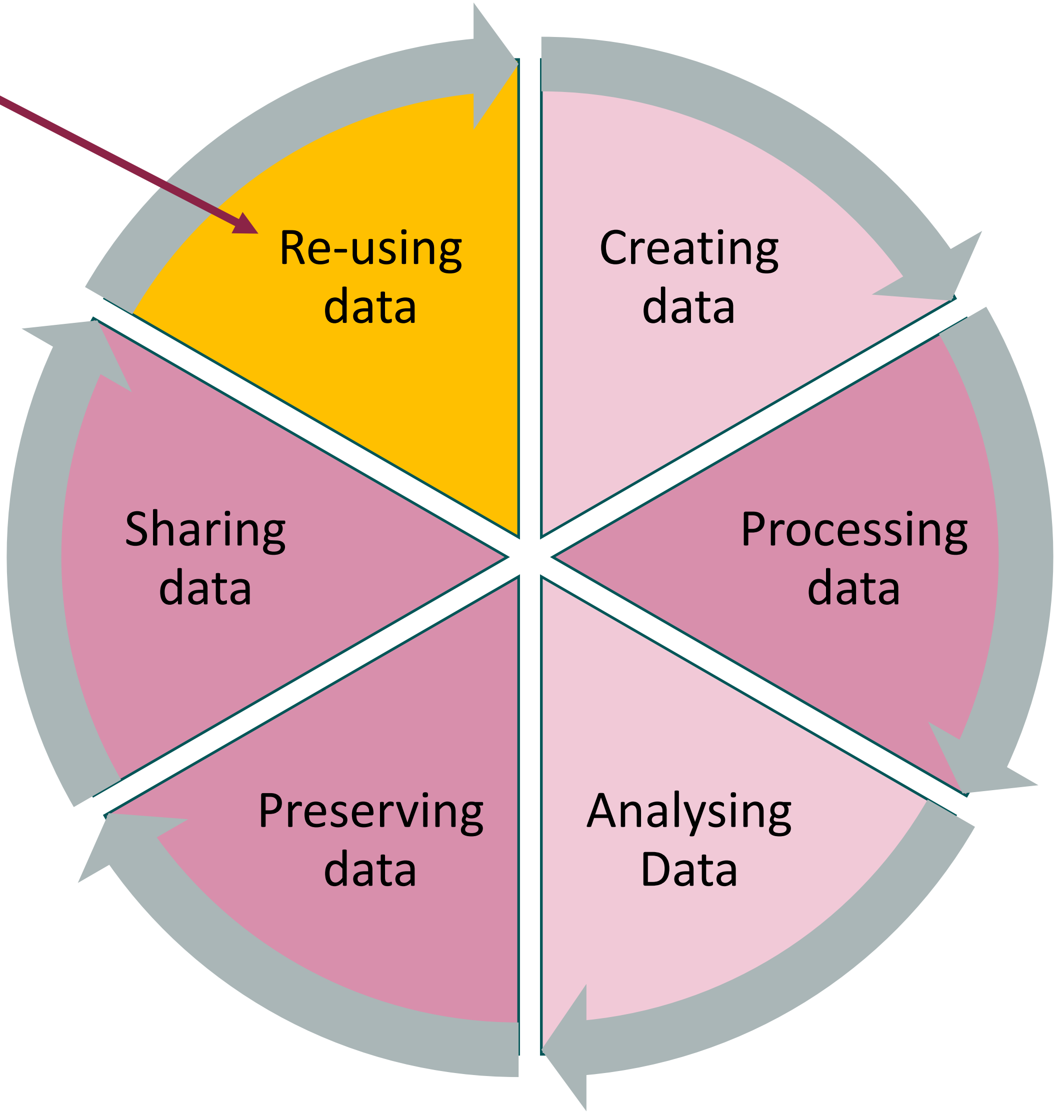


Data Lifecycle

A dataset has a longer lifespan than the research project that creates it

Data can be used and re-used for future research, if:

- shared
- managed well
- properly preserved
- made available



Research Data Management: organisation, storage and preservation of data created during a research project

Which data?

Research data can take many different forms, but is essentially the **evidence used to inform or support research conclusions.**

- Video and voice recordings
- Questionnaires and interview transcripts
- Test results held in text files and spreadsheets
- Archive materials and handwritten notes
- Code and software
- Photographs and slides
- Laboratory notebooks
- ….

ALL DATA

Why good RDM is necessary?

- Efficiency and ease of data control, with reduced risk of data loss
- Greater visibility of data, leading to increased citations and future collaborations
- Demonstration of research integrity and validation of research results
- Compliance with funders, publishers and institutional policies and expectations
- Greater impact of research through knowledge transfer
- Research advances through reuse of data by researchers around the world

Funders data policies

● Full Coverage ◐ Partial Coverage ○ No Coverage

	Policy Coverage		Policy Stipulations					Support Provided			
Research Funders	Published outputs	Data	Time limits	Data plan	Access/ sharing	Long-term curation	Monitoring	Guidance	Repository	Data centre	Costs
AHRC	●	●	●	●	●	◐	○	●	○	◐	◐
BBSRC	●	●	●	●	●	●	●	●	●	◐	●
CRUK	●	●	●	●	●	●	●	◐	●	○	○
EPSRC	●	●	●	◐	●	●	●	◐	○	○	●
ESRC	●	●	●	●	●	●	●	●	●	●	◐
MRC	●	●	●	●	●	●	○	◐	●	○	◐
NERC	●	●	●	●	●	●	●	●	●	●	◐
STFC	●	●	●	●	●	●	●	◐	●	◐	◐
Wellcome Trust	●	●	●	●	●	●	●	●	●	◐	●

http://www.dcc.ac.uk/resources/policy-and-legal/overview-funders-data-policies

Publishers, University

Publisher requirements

- Some publishers now require that researchers undertake to share their research data.
- Researchers should consult journal data policies directly in order to assess their specific requirements.

University requirements:

- Stellenbosch University research data management regulations 2021 (next planned review: 31/12/2022):

https://www.sun.ac.za/english/Documents/Terms_and_conditions/Current/RDM_Regulations_English.pdf

be FAIR…

Findable
Accessible
Interoperable
Reusable

> ***FAIR*** *Principles for scientific data management: Principles that propose that all scholarly output should be Findable, Accessible, Interoperable, and Reusable.*
>
> *Stellenbosch University research data management regulations 2021, page 4*

The Data Management Plan

A Data Management Plan (DMP) is the document that describes in sufficient details:

- what data will be created;
- how data will be collected;
- how data will be processed to be made ready for analysis;
- how raw and processed data will be stored, preserved;
- how data will be shared, and with which limitations.

Needs a DMP

Needs a DMP!

The Data Management Plan

- Plan data management early
- Assign roles and responsibilities
- Design data management according to needs and purpose of research
- Implement and review data management throughout research

Data Management Plan: content

- Data Collection
- Documentation and Metadata
- Storage and Backup
- Preservation
- Sharing and Reuse
- Responsibilities and Resources
- Ethics and Legal Compliance

DMP: Data Collection

- What types of data will you collect, create, link to, acquire and/or record?
- What file formats will your data be collected in? Will these formats allow for data re-use, sharing and long-term access to the data?
- What conventions and procedures will you use to structure, name and version-control your files to help you and others better understand how your data are organized?

REDCap®

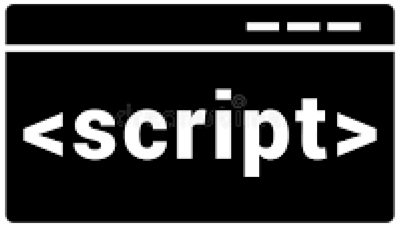
Checking, correcting

Data entry and coding

Missing data

Incongruencies

Documentation or cleaning process and safekeeping of raw data



DMP: Documentation and Metadata

- What documentation will be needed for the data to be read and interpreted correctly in the future?
- How will you make sure that documentation is created or captured consistently throughout your project?
- If you are using a metadata standard and/or tools to document and describe your data, please list here.

A codebook

1) short description of the study design, purpose, methodology, the survey process, instruments, findings; and records of survey team members;
2) description of the population, and any special conditions associated with that population, such as groups that were oversampled include information concerning the measurement class of the variables;
3) outline the way in which data is constructed;
4) all variable names, coding and categorisation;
5) accurate reproduction of each questionnaire item.

Manually generated

Software generated

Example 2

```
. codebook pid w1_a_outcome w1_a_hl30fl w1_a_hl30fev w1_a_hl30pc w1_a_hl30cb

------------------------------------------------------------------------------------------------
pid                                                                            Person identifier
------------------------------------------------------------------------------------------------

                  type:  numeric (long)

                 range:  [301011,799992]              units:  1
         unique values:  59,677                   missing .:  0/59,677

                  mean:    550287
             std. dev:     186349

           percentiles:        10%       25%       50%       75%       90%
                            309236    322668    620408    727458    758853

------------------------------------------------------------------------------------------------
w1_a_outcome                                                                   Interview outcome
------------------------------------------------------------------------------------------------

                  type:  numeric (byte)
                 label:  w1_outcome

                 range:  [1,2]                        units:  1
         unique values:  2                        missing .:  42,805/59,677

            tabulation:  Freq.   Numeric  Label
                        15,631         1  Successfully Interviewed
                         1,241         2  Refused/ Not Available
                        42,805         .

------------------------------------------------------------------------------------------------
w1_a_hl30fl                   j2_1 - In the last 30 days respondent experienced Flu symptoms?
------------------------------------------------------------------------------------------------

                  type:  numeric (byte)
                 label:  w1_yn

                 range:  [1,2]                        units:  1
         unique values:  2                        missing .:  44,133/59,677

            tabulation:  Freq.   Numeric  Label
                         3,767         1  Yes
                        11,777         2  No
                        44,133         .
```

DMP: Storage and Backup

- What are the anticipated storage requirements for your project, in terms of storage space (in megabytes, gigabytes, terabytes, etc.) and the length of time you will be storing it?
- How and where will your data be stored and backed up during your research project?
- How will the research team and other collaborators access, modify, and contribute data throughout the project?

DMP: Preservation

- Where will you deposit your data for long-term preservation and access at the end of your research project?
- Indicate how you will ensure your data is preservation ready. Consider preservation-friendly file formats, ensuring file integrity, anonymization and de-identification, inclusion of supporting documentation.

DMP: Sharing and Reuse

- What data will you be sharing and in what form? (e.g. raw, processed, analyzed, final).
- Have you considered what type of end-user license to include with your data?
- What steps will be taken to help the research community know that your data exists?

DMP: Responsibilities and Resources

- Identify who will be responsible for managing this project's data during and after the project and the major data management tasks for which they will be responsible.
- How will responsibilities for managing data activities be handled if substantive changes happen in the personnel overseeing the project's data, including a change of Principal Investigator?
- What resources will you require to implement your data management plan? What do you estimate the overall cost for data management to be?

DMP: Ethics and Legal Compliance

- If your research project includes sensitive data, how will you ensure that it is securely managed and accessible only to approved members of the project?
- If applicable, what strategies will you undertake to address secondary uses of sensitive data?
- How will you manage legal, ethical, and intellectual property issues?

Guidelines & Resources

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RDM services @ Library & Information Service

Advisory services

Consultation sessions relating to the following issues:

- Creation of data management plans
- Management of research data during the research process
- Management of research data after the conclusion of the research process
- Data sharing and dissemination methods
- Appropriate mediums for sharing and disseminating research data

Training sessions

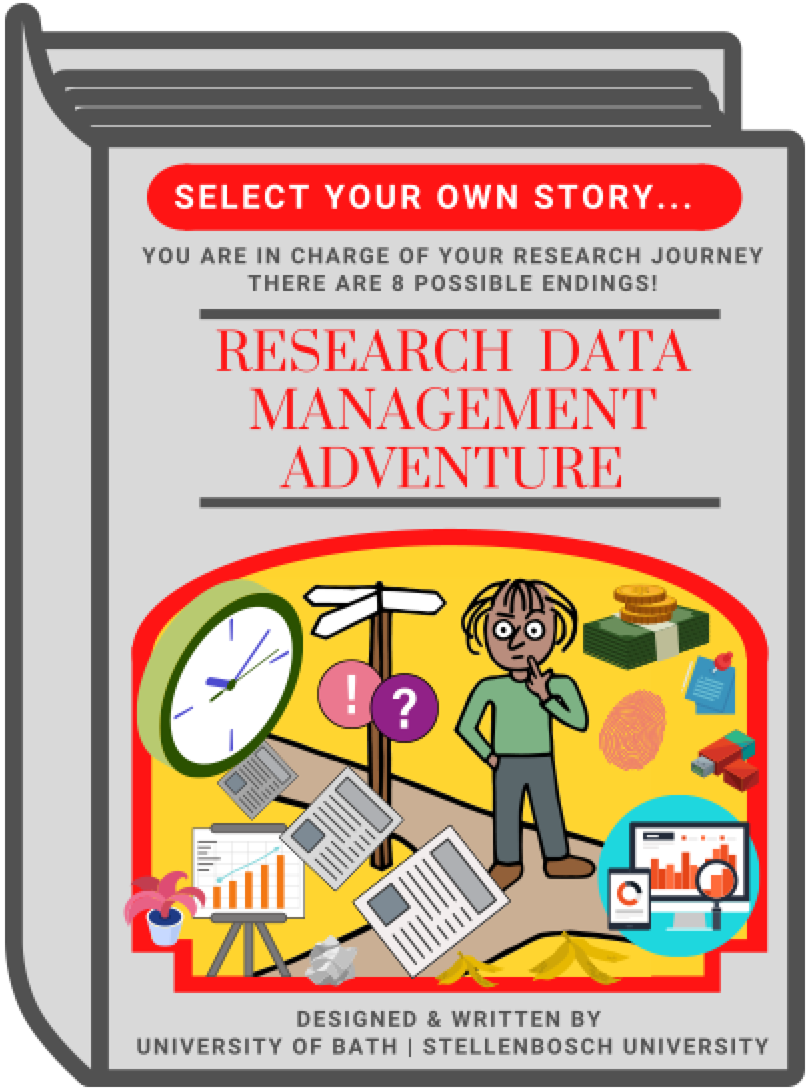
Scheduled training sessions

The following types of sessions are currently offered at the moment:

- Overview of research data management
- Data sharing and dissemination

https://library.sun.ac.za/en-za/Research/rdm/Pages/default.aspx

https://www.sun.ac.za/english/Documents/Terms_and_conditions/Current/RDM_Regulations_English.pdf



https://rdm-games.gitlab.io/rdm-adventure/

Guidelines and resources: Costing

UK Data Service - Data management costing tool and checklist

The UK Data Service has prepared this costing tool and checklist to help formulate research data management costs in advance of research starting, for example for inclusion in a data management plan or in preparation for a funding application.

This tool considers the additional costs - above standard planned research procedures and practices - that are needed to preserve research data and make them shareable beyond the primary research team. The checklist indicates the activities to consider and cost to enable good data management. Such additional activities may require extra researcher or administrative staff time input, equipment, software, infrastructure or tools.

There are no hard and fast rules for costing data sharing requirements, as some research projects will pay more attention to detailed data documentation, organisation and formatting than others as part of routine fieldwork or preparation before analysis. Much also depends on the long-term storage, preservation and publication plans beyond the duration of the research itself. When data are deposited with a professional data centre or repository, such as the UK Data Archive, data preservation and dissemination activities are covered by the data centre/repository.

https://dam.ukdataservice.ac.uk/media/622368/costingtool.pdf

Guidelines & Resources

Digital Research Alliance of Canada | Alliance de recherche numérique du Canada

DMP Assistant

https://alliancecan.ca/en/services/research-data-management/dmp-assistant

The DMP Assistant is a national, online, bilingual data management planning tool developed by the Digital Research Alliance of Canada (the Alliance) in collaboration with host institution University of Alberta to assist researchers in preparing data management plans (DMPs). This tool is freely available to all researchers and develops a DMP through a series of key data management questions, supported by best-practice guidance and examples.

Summary questions

- What is data management?
- What does it mean ‘cleaning data”?
- Main steps in data cleaning procedures
- What is a data management plan, and why is necessary?
- What a data management plan should include?

References

UK Data Service. 2013. ‘UK Data Service - Data Management Costing Tool and Checklist’. UK Data Archive, University of Essex. https://dam.ukdataservice.ac.uk/media/622368/costingtool.pdf.

Stellenbosch University. 2021. ‘Stellenbosch University Research Data Management Regulations’. Division of Research Development & LIbrary Information Services, Stellenbosch University. https://www.sun.ac.za/english/Documents/Terms_and_conditions/Current/RDM_Regulations_English.pdf.

‘Research Data Management Overview’. n.d. *Open Access and Open Data Steering Committee* (blog). Accessed 23 June 2022. https://www.library.yorku.ca/web/open/overview/.

Fowler Jr, Floyd J. 2013. *Survey Research Methods*. Sage publications. Chapter 9.

Other resources

‘Research Data Management Adventure’. n.d. Accessed 23 June 2022. https://rdm-games.gitlab.io/rdm-adventure/.

‘DMP Assistant’. n.d. Digital Research Alliance of Canada. Accessed 23 June 2022. https://alliancecan.ca/en/services/research-data-management/dmp-assistant.

Thank you

Annibale Cois

Education Building, 4th floor, Room 4060B

acois@sun.ac.za